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Structure of Mathematical Proofs



Learning Objectives

At the end of the lesson, the students must be able to:

- Explain the importance of proofs in establishing the validity of mathematical statements.
- Identify and differentiate between common types of proofs such as direct proofs, proof by contradiction, proof by contrapositive, and proof of induction.
- Apply proof of contradiction to demonstrate the falsity of an assumption and thereby prove the correctness of a proposition



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Why is proofs important in establishing validity of mathematical statement?

According to Bleiler-Baxter & Pair, for a mathematician, a proof serves to convince or justify that a certain statement is true. But it also helps to increase the understanding of the result and the related concepts. That is why a proof also has the role of explanation.



Direct Proof

It is the most straightforward method of proving a mathematical statement. In this type of proof, you begin with known facts (axiom, definitions, or previously established theorems) and use logical steps to arrive at the statement you're trying to prove.

Structure:

- Assume that the premise (hypothesis) of the statement is true.
- Use logical reasoning and deductions based on the assumption and known facts to show that the conclusion



Example:

Prove that the sum of two even integers is always even.

- Let the two even integers be $a = 2k$ and $b = 2m$, where k and m are integers.
- Their sum is $a + b = 2k + 2m = 2(k + m)$
- Since $k + m$ is an integer, $2(k + m)$ is even. Therefore the sum of two even integers is even.



Proof by Contradiction

It involves assuming that the statement you are trying to prove is false and then showing that this assumption leads to a contradiction. When the assumption results in an impossible situation, you conclude that the original statement must be true.

Structure:

- Assume the opposite (negation) of what you are trying to prove.
- Show that this assumption leads to a logical contradiction (i.e., something that is clearly false or contradictory to known facts).



Example

Prove that square root of 2 is irrational

- Assume that the opposite, i.e., square root of 2 is rational. This means it can be written as a/b , where a and b are integers with no common factors, and b is not equal to 0.
- Then square root of 2 is equal to a/b implies $2 = a^2/b^2$, or a^2 is equal to $2b^2$.
- This implies that a^2 is even, so a must be even. Let $a = 2k$.
- Substituting into the equation gives $(2k)^2 = 2b^2$, or $4k^2 = 2b^2$, which simplifies to $2k^2 = b^2$, meaning b^2 is also even.



Proof by Contrapositive

In a proof of contrapositive, you prove the contrapositive of the statement instead of the statement itself. The contrapositive of a statement "If P, then Q" is "if not Q, then not P". These two statements are logically equivalent, so proving one proves the other.

Structure:

- Instead of proving "if P, then Q" directly, prove its contrapositive: "if not Q, then not P"
- Use logical reasoning to show that if the conclusion is



Example

Prove that if a number is not even, then it is not divisible by 4.

- The contrapositive is: if a number is divisible by 4, then it is even.
- Let n be a number divisible by 4, so $n = 4k$ for some integer k .
- Since $n = 4k = 2(2k)$, n is clearly even because it is divisible by 2.
- Thus, the contrapositive is true, and by logical equivalence, the original statement is also true: if a number



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REFERENCE:

<https://opensa-server.cs.vt.edu/ODSA/Books/Everything/html/Proofs.html>